

András Gémes

shadowshell.io | github.com/gemesa | linkedin.com/gemesa | gemesa@protonmail.com

Summary

Security researcher with 8 years of experience. [Hands-on experience](#) in mobile application security, reverse engineering, malware analysis and vulnerability research. Certified in [Sec+](#), [CASP+/SecX](#), [CEH](#) and [others](#). LLVM compiler engineering (code obfuscation) and embedded systems background. Open to roles in security research.

Work experience

Application Security Researcher @ SGS Brightsight - Delft, Netherlands **June 2026 – Present**

- Analyzing mobile payment applications for vulnerabilities (cryptography, runtime protection, data protection, etc.)
- Evaluating mobile payment applications against the PCI MPoC standard
- Reverse engineering app binaries (Ghidra, Frida) to identify logic flaws and develop PoC exploits

Security Researcher @ HighTec EDV-Systeme GmbH - Budapest, Hungary **Feb 2023 – June 2026**

- Evaluated the security impact of reported LLVM vulnerabilities as a [member of the LLVM security group](#)
- Analyzed Windows, Linux and Android [malware](#), writing [detection rules](#) and config extractors ([Ghidra](#), [Python](#))
- Implemented custom [LLVM-based C++ code obfuscation pass plugins](#) based on malware research

Application Security Engineer @ Knorr-Bremse R&D Center - Budapest, Hungary **May 2018 – Jan 2023**

- Implemented static (Clang-Tidy) and dynamic (Valgrind) application security testing across embedded C codebases
- Used AFL++ fuzzing and code review to discover embedded systems memory corruption vulnerabilities and build PoC exploits
- Built iOS and Android apps for real-time vehicle data visualization

Technical skills

Reverse engineering (static): DiE, Ghidra, IDA, ILSpy, dnSpy, Binwalk, Apktool, jadx, otool, ipsw

Reverse engineering (dynamic): x64dbg, Sysinternals, LLDB, GDB, Frida, ADB, QEMU, Qiling, VirtualBox, strace

Detection engineering: YARA, Sigma, Suricata, capa

Programming languages: C, C++, Rust, Objective-C, Swift, Python 3, Java, Assembly (ARM64, x86-64), Bash

Vulnerability research: AFL++, checksec, ROPgadget, Clang-Tidy, Valgrind

Network and traffic analysis: Wireshark, Burp Suite, mitmproxy, Zeek, FakeNet-NG, INetSim

Platforms and DevOps tools: Linux (Fedora, Ubuntu), macOS, Windows, Git, Docker, GitHub Actions, Jenkins

Certifications

[CompTIA Security+](#), [CompTIA CASP+/SecurityX](#), [EC-Council CEH](#), [TCM Security PMAT](#), [Invoke RE IMBT](#) and [others](#).

Open source contributions

- [ghidra](#): contributing bug reports and patches to Ghidra, focusing on the BSim, Debugger and FunctionID features
- [threat-detection-rules](#): publishing YARA, Sigma and Suricata rules for analyzed malware families
- [ghidra-scripts](#) | [reversing-scripts](#): developing custom scripts to support reverse engineering
- [phantom-pass](#) | [o-mvll](#): implementing and improving LLVM-based code obfuscation pass plugins
- [rust-arm64](#): writing a Rust book about analyzing Rust-to-ARM64 compilation

Education

MSc in Mechatronics Engineering @ [Budapest University of Technology and Economics](#) **Feb 2016 – June 2018**

BSc in Mechatronics Engineering @ [University of Pannonia](#) **Sept 2012 – Jan 2016**